

Every configuration against plain lifelong PIBT

Blue beats plain PIBT on that map; red loses to it. Percentage difference in throughput, with a two-sided permutation test over the same seeds; bold is $p < 0.05$.

SPAR (full)	p = 0.10 +23%	p = 0.49 -7%	p < 0.05 -45%	p < 0.05 -38%
SPAR (aisle direction)	p = 0.06 +27%	p = 0.06 +21%	p < 0.05 -18%	p < 0.05 -19%
SPAR (aisle managed)	p = 0.06 +27%	p < 0.05 -28%	p < 0.05 -57%	p < 0.05 -57%
PIBT + hysteresis	p = 0.06 +27%	p = 0.19 +13%	p < 0.05 -27%	p < 0.05 -33%
PIBT + robot direction	p = 0.11 +21%	p < 0.05 +20%	p < 0.05 -47%	p < 0.05 -44%
PIBT + turning cost	p < 0.05 +28%	p < 0.05 +50%	p < 0.05 -24%	p < 0.05 -18%
PIBT + entry admission	p = 0.06 +27%	p = 0.57 -5%	p < 0.05 -58%	p < 0.05 -48%
aisle direction, enforced	p = 0.06 +27%	p < 0.05 -35%	p < 0.05 -78%	p < 0.05 -68%
recovery, uncorroborated	p = 0.06 +26%	p < 0.05 -83%	p < 0.05 -62%	p < 0.05 -65%
	bottleneck aisle-constrained	corridors aisle-constrained	narrow open floor	medium open floor